

Homework (Jalapeno)

- Q1.** Solve the inequality $2x - 5 > 3$.
(A) $x > 4$
(B) $x < 4$
(C) $x > -4$
(D) $x < -4$
(E) $x = 4$
- Q2.** Graph the solution to $x + 2 \leq 7$ on a number line.
(A) $(-\infty, 5]$
(B) $[5, \infty)$
(C) $(-\infty, 5)$
(D) $(-5, \infty)$
(E) $[5, 7]$
- Q3.** The ancient Egyptians built the Great Pyramid with a height of 146.5 meters. If the height h of the pyramid is reduced by x meters, the inequality $h - x > 120$ represents the condition that the reduced height is greater than 120 meters. Solve for x .
(A) $x < 26.5$
(B) $x > 26.5$
(C) $x = 26.5$
(D) $x < -26.5$
(E) $x > -26.5$
- Q4.** The Babylonians used a sexagesimal (base-60) system for mathematics. If x is a number in this system and $x + 20 > 40$, solve for x .
(A) $x > 20$
(B) $x < 20$
(C) $x = 20$
(D) $x > 40$
- (E) $x < 40$
- Q5.** The ancient Greeks used the method of exhaustion to calculate areas. If x is the area of a shape and $x - 3 \geq 5$, solve for x .
(A) $x \geq 8$
(B) $x \leq 8$
(C) $x = 8$
(D) $x > 8$
(E) $x < 8$
- Q6.** Write the solution to $-3x + 2 > 5$ in interval notation.
(A) $(-\infty, -1)$
(B) $(-1, \infty)$
(C) $(-\infty, 1)$
(D) $(-1, 1)$
(E) $(-1, \infty)$
- Q7.** Solve the inequality $\frac{2x+3}{5} \leq 2$.
(A) $x \leq \frac{7}{2}$
(B) $x \geq \frac{7}{2}$
(C) $x = \frac{7}{2}$
(D) $x > \frac{7}{2}$
(E) $x < \frac{7}{2}$
- Q8.** The ancient Chinese used abacuses for calculations. If x is a number on an abacus and $-2x + 1 > 7$, solve for x .
(A) $x < -3$
(B) $x > -3$
(C) $x = -3$
(D) $x < 3$
(E) $x > 3$

Open Ended Questions (FRQ)

- Q9.** Solve the inequality $x^2 - 4x - 3 > 0$ and graph the solution on a number line. Write your answer in interval notation.
- Q10.** The ancient Romans built a bridge with an arch that had a height of 15 meters. If the height h of the bridge is reduced by x meters, the inequality $h - x > 10$ represents the condition that the reduced height is greater than 10 meters. Solve for x , graph the solution on a number line, and write your answer in interval notation.

Chef's Tips

- Tip 1: Emphasize the importance of understanding the inequality sign and its direction.
- Tip 2: Use number lines to visualize the solution to inequalities.
- Tip 3: Encourage students to check their solutions by plugging values back into the original inequality.